Aishwarya Kandasamy

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SUMMARY

Lead Software Engineer with 5+ years of experience designing and operating large-scale distributed systems, data pipelines, and cloud-native platforms. Experienced in building event-driven architectures using AWS serverless services, Kafka-style streaming systems, and scalable backend services in Python, Java, and Node.js. Proven track record delivering high-throughput ingestion platforms, real-time analytics pipelines, and production systems handling tens of thousands of events per hour. Skilled in system design, reliability engineering, and translating complex product requirements into scalable production systems.

SKILLS

Languages: Java, Python, JavaScript, TypeScript, Node.js, SQL, C, C++
Frameworks: React.js, Redux, Spring Boot, Next.js, Express.js, HTML5, CSS
Cloud & Infras: AWS (Lambda, EC2, S3, DynamoDB, API Gateway, IAM, VPC, SQS, SNS, EventBridge, CloudWatch), Terraform, CloudFormation, Docker, Kubernetes
Data & Streaming: Snowflake, Apache Kafka, PostgreSQL, MongoDB, Data Pipelines, ETL/ELT
DevOps & Tools: Git, GitHub, Bitbucket, Jenkins, Jira, Swagger, Postman, Grafana, GitHub Copilot, Cursor AI, CI/CD Pipelines
Core Engineering: Distributed Systems, Microservices Architecture, Event-Driven Architecture, System Design, API Design, OOP, Data Structures, Algorithms, Design Patterns, Monitoring & Observability, SDLC, Micro-Frontends, High Availability Systems, Performance Optimization, Frontend Design Patterns, Build Systems, Dependency Management
APIs & Protocols: REST, GraphQL, JSON
Operating System: Linux

WORK EXPERIENCE

Utilyze, Inc Sep 2025 - Present
Lead Software Engineer

- Architected and owned an **event-driven** ingestion platform using **Python** and **Node.js** on **AWS Lambda, SQS**, and **S3**, processing **80K+** meter readings/hour with idempotent consumers, retry policies, and dead-letter queues for fault-tolerant real-time analytics.
- Developed scripts in **Python** to automate operational workflows, deployment tasks, and system monitoring.
- Implemented **CI/CD pipelines** and containerized build workflows using **Docker** and **Jenkins**, improving release reliability and reducing deployment failures across multiple environments.
- Built a synthetic **data generation framework** using temperature-based scaling and appliance-specific usage patterns to simulate realistic scenarios for leak detection and appliance **classification model training**.
- Delivered **customer-facing analytics features** including usage insights, leak alerts, and peak demand trends by defining **API contracts** and collaborating cross-functionally to ship **production-ready systems and demos**.
- Designed the end-to-end data architecture including ingestion (**SQS/Kafka**-style streaming), storage (**S3 data lake**), and analytics layer (**Snowflake**), enabling reliable real-time and batch analytics workflows.
- Defined service-level objectives (**SLOs**), observability dashboards, and alerting metrics using **CloudWatch** and **Grafana** to monitor pipeline reliability and ingestion lag.
- Mentored junior engineers through **code reviews**, **architecture discussions**, and technical guidance to improve development velocity and engineering quality.

Open Avenues Jun 2025 - Aug 2025
Software Engineer Consultant

- Developed a production-ready **Graph UI** library using **React** and **TypeScript** to visualize large-scale relational schemas in real-time.
- Built lightweight backend services using **Node.js** to expose **RESTful APIs**, handle graph configuration requests, and support real-time updates between client and server.
- Implemented efficient data transformation pipelines to convert database metadata into **dynamic graph** structures, enabling scalable visualization of relational datasets.
- Developed reusable **UI components** and integrated **charting/graph libraries**, improving maintainability and accelerating feature delivery across visualization modules.
- Collaborated in an **Agile Scrum** environment using **Git** and **Jira**, participating in sprint planning, code reviews, and weekly product demos.

Illinois Institute of Technology Jan 2024 - May 2024
Software Engineer Practicum

- Developed a scalable **NLP processing pipeline** in **Python** to classify **30K+** academic articles using BERT-based multi-label classification models.
- Built automated preprocessing **pipelines** for text normalization, tokenization, and feature extraction using Hugging Face and Keras.

- Implemented data balancing and training pipelines using Dynamic ML SMOTE to address class imbalance and improve model robustness across multiple labels.
- Designed evaluation and benchmarking workflows using precision, recall, F1-score, and cross-validation to compare multiple models including TF-IDF, SVM, Rakel ensembles, and BERT.

Praxio Aug 2022 - Jul 2023

Software Engineer

- Developed scalable **Java Spring Boot** microservices for task turnaround analytics processing **50K+** tasks/month, enabling horizontally scalable distributed services.
- Configured **Docker** and **Kubernetes** environments to support automated builds, testing, and deployment workflows across development and QA environments.
- Implemented **monitoring** and **observability** using **Prometheus** and **Grafana**, enabling real-time system metrics tracking and faster production issue detection.
- Collaborated with product managers and design teams to deliver production features while maintaining system performance and reliability in a **multi-tenant SaaS** environment.
- Automated **AWS** provisioning with **Terraform**, cutting deployment errors by 30% and enabling feature delivery for 50K+ tasks.

Athenahealth Jan 2020 - Jul 2022

Software Engineer

- Led full-stack development using **React** and **Node.js** for a high-concurrency clinical workflow platform supporting **10K+** concurrent requests.
- Designed **CI/CD pipelines** with **Blue-Green deployment** strategies using **Jenkins**, enabling automated build, test, and release processes.
- Optimized performance of complex queries across **PostgreSQL** in a multi-tenant **SaaS** environment, reducing query time by **50%**.
- Hardened **REST APIs** for high-concurrency workloads (**10K+** requests) using timeout, retry, and exponential backoff strategies, reducing failure rate by **25%** and improving service stability under **peak traffic**.

EDUCATION

Illinois Institute of Technology Aug 2023 - May 2025

Master of Science in Computer Science and Data Science

Velammal Engineering College Jun 2016 - Apr 2020

Bachelor of Engineering, Computer Science

CERTIFICATION

AWS Certified Solution Architect - Associate